BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 283 · TCP/IP stack, protocols and DNS

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which device typically forwards packets between different IP networks?

Q2. What does an edge represent in a network graph?

Q3. How many bits make one byte?

Q4. A sender and receiver are on different networks. Can a switch alone normally route IP packets between them?

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I Do · Read new knowledge

Key vocabulary

application layer — Protocols offering services to applications, including DNS and HTTP.

transport layer — End-to-end transport protocols such as TCP and UDP.

internet layer — Network addressing and forwarding using IP.

link layer — Local transfer of frames over media such as Ethernet or Wi-Fi.

DNS — Domain Name System; supports name-to-address resolution.

New learning

Start with a hostname — The browser uses DNS (an application-layer protocol) to resolve school.example to an IP address. A cached result may avoid further queries.

Understand four layers — Application supports services such as HTTP(S), DNS and SMTP; transport includes TCP and UDP; internet uses IP; link transfers frames on the local connection.

Build an HTTPS connection — A browser normally establishes TCP to port 443, negotiates TLS, then exchanges encrypted HTTP requests and responses. DNS lookup happens first if the IP is unknown.

Worked example

A user opens https://library.example. Explain what DNS, TCP and HTTPS each do.

1. Name resolution — The resolver returns the IP address for library.example (or uses a cached answer).

2. Transport connection — TCP establishes a connection to the server, typically destination port 443.

3. Application exchange — TLS secures the connection; HTTPS carries the page request and response.

We Do · Practise with guidance

A pupil sends email using SMTP. Place SMTP, TCP and IP into the correct TCP/IP layers.

Identify the application

SMTP is an application-layer protocol for transferring outgoing email.

Identify the other layers

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

TCP provides reliable transport for SMTP; IP handles inter-network addressing and routing; link delivers frames on each local link.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Which TCP/IP layer contains DNS, and what does DNS do? [2]

Question 2

Place TCP, IP, and HTTP into the correct layers and explain one role of each. [6]

Question 3

A pupil says DNS encrypts webpage traffic and TCP chooses the Wi-Fi channel. Correct both errors, identifying the relevant protocols/layers. [4]

Exit Ticket · Answer before checking

DNS and HTTPS are both application-layer protocols in the TCP/IP model.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Router.

Q2. A connection between two vertices.

Q3. Eight.

Q4. No. A router (or routing-capable layer-3 switch) is needed to route between IP networks.

You Do pages · Independent questions

Q1. [2] Application layer [1]. DNS resolves domain names to IP addresses (among other DNS record types) [1].

Q2. [6] HTTP: application layer [1], handles web requests/responses [1]; TCP: transport layer [1], delivers an ordered reliable byte stream [1]; IP: internet layer [1], provides packet addressing and routing across networks [1].

Q3. [4] DNS performs name resolution, not website-content encryption [1]; TLS provides encryption for HTTPS traffic [1]; TCP is a transport protocol for reliable ordered data [1]; wireless channel selection is a Wi-Fi/link-layer concern [1].

Exit Ticket · Model answer

True, though they perform different functions.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding